Bilinear Modelling of Monthly Sea Surface Temperature in the Niño 3 Region with a Fourier Seasonal Component

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Abstract

The El Niño Southern Oscillation (ENSO) produces asymmetric sea surface temperature (SST) anomalies in the eastern equatorial Pacific, with El Niño events tending to be stronger than La Niña events. Such asymmetry cannot be explained by a linear model with Gaussian errors, which motivates the use of nonlinear models. This study aims to model and forecast the monthly SST of the Niño 3 region from January 1950 to June 2026 using a bilinear model after removing the seasonal component with a Fourier series. The best seasonal component according to the BIC consists of two harmonics and a linear trend. The Lagrange multiplier test rejects the linearity hypothesis at the 5% significance level. A total of 832 BL(p, r, m, k) models were evaluated using the rolling-origin RMSE on the training data, and the BL(2,2,1,1) model was selected as the model with the smallest RMSE. On the test data, the one-step-ahead forecast RMSE of the BL(2,2,1,1) model is 0.258 °C, which is smaller than that of the ARMA(2,2) model, the persistence method, and the climatology method, and comparable to that of the AR(5) model. In El Niño months, the BL(2,2,1,1) model yields the smallest RMSE among all benchmark models. The forecast based on the full data shows a positive anomaly that peaks at 1.97 °C in August 2026 and then weakens to 1.55 °C in December 2026.

Keywords *Bilinear Model, ENSO, Forecasting, Sea Surface Temperature*

1 Introduction

The El Niño Southern Oscillation (ENSO) is the strongest year-to-year fluctuation of the global climate system and affects global climate, marine and terrestrial ecosystems, fisheries, and human activities (Timmermann et al. 2018). ENSO conditions are commonly monitored through the sea surface temperature (SST) of the equatorial Pacific, including the Niño 3 region located at 5°N to 5°S and 90°W to 150°W (Trenberth 1997). Burgers and Stephenson (1999) showed that SST anomalies in the eastern Pacific have a positive skewness that is related to the predominance of El Niño events over La Niña events. If ENSO were generated by a linear system driven by Gaussian noise, SST anomalies would be normally distributed (Burgers and Stephenson 1999). The observed asymmetry therefore indicates the presence of nonlinear processes.

One class of nonlinear time series models is the bilinear model introduced by Granger and Andersen (1978a). This model extends the ARMA model by adding product terms between

past values of the series and past errors. Subba Rao (1981) derived sufficient conditions for stationarity, expressions for the variance and covariance, and conditions for invertibility of bilinear models, and applied the model to sunspot numbers. Gabr and Subba Rao (1981) developed the estimation and forecasting of subset bilinear models and compared their forecasts with those of linear AR and threshold AR models.

This study aims to model the monthly SST of the Niño 3 region using a bilinear model with a Fourier seasonal component. The model order is identified in successive stages, model selection is carried out with rolling-origin evaluation, and the forecast accuracy on the test data is compared with that of linear models and two simple benchmark methods. The selected model is then used to forecast the SST from July to December 2026.

2 Methods

2.1 Data

The data used in this study are the monthly SST of the Niño 3 region from January 1950 to June 2026, consisting of 918 observations obtained from the NOAA Climate Prediction Center. The data are derived from the Extended Reconstructed Sea Surface Temperature version 5 (ERSSTv5), a monthly global SST dataset with a resolution of $2^\circ \times 2^\circ$ (Huang et al. 2017). The variable used is the actual SST rather than the anomaly, so that the seasonal component can be estimated from the training data only. The data were split sequentially into training data of 780 observations (85%, January 1950 to December 2014) and test data of 138 observations (15%, January 2015 to June 2026). Order identification, parameter estimation, and model selection use the training data only.

2.2 Seasonal component

The SST series X_t is decomposed into a seasonal component S_t and a stochastic component Y_t as follows

$$X_t = S_t + Y_t, \qquad S_t = \mu + \delta \frac{t}{120} + \sum_{q=1}^{K} \left[\beta_q \sin\left(\frac{2\pi q t}{12}\right) + \lambda_q \cos\left(\frac{2\pi q t}{12}\right) \right], \tag{1}$$

where $t = 0, 1, \ldots$ denotes the month index starting from January 1950, so that δ is the trend in °C per decade. Fourier series can represent seasonal patterns, including complex seasonal patterns (De Livera et al. 2011). The parameters of the seasonal component are estimated by least squares. The number of harmonics $K \in \{1, \ldots, 6\}$ and the inclusion of the trend are determined by the smallest BIC value. This criterion imposes a penalty of $\frac{1}{2} k \log n$ on the log-likelihood and therefore tends to select models of lower dimension than the Akaike criterion (Schwarz 1978). The seasonal component for the test period and the forecast period is computed from equation (1) with the same parameters.

2.3 Bilinear model

The bilinear model BL(p, r, m, k) for the series Y_t is defined as (Granger and Andersen 1978a, Subba Rao 1981, Subba Rao and Gabr 1984)

$$Y_t + \sum_{u=1}^{p} a_u Y_{t-u} = \alpha + \sum_{l=1}^{r} c_l \, e_{t-l} + \sum_{i=1}^{m} \sum_{j=1}^{k} b_{ij} \, Y_{t-i} \, e_{t-j} + e_t, \tag{2}$$

where a_u, c_l, and b_{ij} are the coefficients of the autoregressive (AR), moving average (MA), and bilinear parts, respectively, α is a constant, and e_t are independent errors with zero mean and variance σ^2. The AR coefficients are written on the left-hand side, so that $a_u = -\phi_u$ in the usual ARMA notation. If all b_{ij} are zero, equation (2) reduces to an ARMA(p, r) model. Gabr and Subba Rao (1981) discussed the subset form, which contains only selected lags in each part of the model.

The MA and bilinear terms with the same error lag can be written as

$$\Big(c_l + \sum_{i=1}^{m} b_{il} \, Y_{t-i} \Big) \, e_{t-l}.$$

This form shows that the coefficient of a past error depends on previous values of the series, so that the bilinear model can generate asymmetric behaviour.

Granger and Andersen (1978b) proposed a more general definition of invertibility and applied it to linear, nonlinear, and bilinear models. A model is invertible if the errors can be reconstructed from the observed data and the effect of incorrect initial errors vanishes over time. In this study, invertibility is checked numerically by propagating a perturbation $d_t = 1$ for $t \le \gamma$ through the recursion

$$d_t = -\sum_{l=1}^{r} c_l \, d_{t-l} - \sum_{i=1}^{m} \sum_{j=1}^{k} b_{ij} \, Y_{t-i} \, d_{t-j}, \qquad t > \gamma. \tag{3}$$

The model is declared invertible if $\log_{10} |d_t| < 0$ at the end of the data period.

2.4 Parameter estimation

Parameter estimation for bilinear models was discussed by Subba Rao (1981) and Gabr and Subba Rao (1981). In this study, the parameters $\boldsymbol{\theta} = (a, c, b, \alpha)$ are estimated by conditional least squares. The errors are computed recursively from equation (2) with initial values $e_t = 0$ for $t \le \gamma$, and the sum of squared errors

$$Q(\boldsymbol{\theta}) = \sum_{t=\gamma+1}^{N} e_t^2(\boldsymbol{\theta}) \tag{4}$$

is minimised using Gauss-Newton iterations. The derivatives $\partial e_t / \partial \boldsymbol{\theta}$ are computed recursively, and each iteration uses step halving to guarantee a decrease in Q. The value $\gamma = 12$ is used for all models, so that every model is estimated on the same $n = N - \gamma$ observations and the AIC values are comparable across models. The AIC is computed as $n \log \hat{\sigma}^2 + 2q$, where q is the

number of parameters (Akaike 1974). The iterations are limited to 500 and, if convergence has not been reached, are continued from the last estimate for at most 2000 further iterations.

2.5 Order identification

The stationarity of Y_t is tested with the Augmented Dickey-Fuller (ADF) test, whose null hypothesis is that the series has a unit root (Dickey and Fuller 1979, Said and Dickey 1984), and with the KPSS test, whose null hypothesis is that the series is stationary (Kwiatkowski et al. 1992).

The order of the linear part is identified from the autocorrelation function (ACF), the partial autocorrelation function (PACF), and the AIC of AR models. Kim (1992) showed that the ACF of a bilinear model $\mathrm{BL}(p, q, r, s)$ has a pattern similar to the ACF of an $\mathrm{ARMA}(p, l)$ model with $l = \max(q, s)$. Hence, the ACF and PACF cannot distinguish the MA part from the bilinear part, and Kim (1992) recommended selecting the model among several candidates using the AIC. In this study, the ACF and PACF are therefore used to determine the candidate orders, while the final model is determined by the criterion in Section 2.6.

The presence of bilinear terms is tested with the Lagrange multiplier (LM) test against bilinear alternatives (Saikkonen and Luukkonen 1988). The test statistic is obtained from an auxiliary regression in which the residuals of the linear model are regressed on the regressors of the linear model and on the product terms corresponding to the bilinear terms. Teräsvirta (1994) recommended the F version of this statistic for short series because its size is closer to the nominal level. The LM test cannot be constructed against an alternative that contains both $Y_{t-i}e_{t-j}$ and $Y_{t-j}e_{t-i}$, so the bilinear terms are restricted to $i \geq j$ (Bera and Higgins 1993). The relevant lag pairs are identified from the correlation between the AR residuals $\hat{e}_t$ and $(Y_{t-i} - \bar{Y})\hat{e}_{t-j}$ for $i, j = 1, \ldots, 12$, with significance bounds $\pm 1.96/\sqrt{n}$. Other approaches to identifying bilinear terms include the ACF of the squared series (Li 1984), third-order moments (Kumar 1986, Wu and Shih 1992), and a genetic algorithm with the AIC or BIC criterion (Chen et al. 2001).

2.6 Model selection

The candidate models are combinations of the AR order p based on the significant PACF lags, the MA order $r = 0, 1, \ldots, 12$, and a bilinear part containing all coefficients b_{ij} with $i \leq m \leq 5$ and $j \leq k \leq 3$, including models without bilinear terms. Candidates that do not converge or are not invertible are excluded.

Each candidate is evaluated with the rolling-origin method, in which the forecast origin is updated successively to obtain a number of forecast errors (Tashman 2000). Five origins are placed in the last 120 months of the training data, namely December 2004, 2006, 2008, 2010, and 2012. At each origin, the seasonal component and the model are re-estimated using the data up to that origin, and rolling one-step-ahead forecasts are made for the following 24 months. The rolling-origin RMSE is computed from these $5 \times 24 = 120$ forecast errors.

The model with the smallest rolling-origin RMSE among the convergent and invertible candidates is selected as the best model. This criterion is used because the ACF and PACF cannot determine the MA order of a bilinear model unambiguously (Kim 1992).

2.7 Diagnostic checking

Model adequacy is checked with the Ljung-Box test on the residuals (Ljung and Box 1978), the McLeod-Li test, which applies the Ljung-Box statistic to the squared residuals (McLeod and Li 1983), and the skewness test u_1 and kurtosis test u_2 for linear stochastic processes (Lomnicki 1961). The joint statistic $u_1^2 + u_2^2$ is compared with the χ_2^2 distribution. All tests use 12 lags.

2.8 Forecasting

The one-step-ahead forecast of model (2) can be computed exactly as

$$\hat{Y}_{t+1|t} = \alpha - \sum_{u=1}^{p} a_u Y_{t+1-u} + \sum_{l=1}^{r} c_l \, e_{t+1-l} + \sum_{i=1}^{m} \sum_{j=1}^{k} b_{ij} \, Y_{t+1-i} \, e_{t+1-j}. \tag{5}$$

For $h \geq 2$, the terms $b_{ij} Y_{t+h-i} e_{t+h-j}$ with $i \leq j$ contain products of series values and errors that are not independent, so their conditional expectation is not zero. A chained forecast that replaces future errors with zero, hereafter called the naive forecast, ignores the contribution of these terms. Brown and Mariano (1989) studied several predictors for dynamic nonlinear models, namely the deterministic predictor, the Monte Carlo predictor, and the predictor based on the empirical distribution of the residuals. In this study, $E(Y_{t+h} \mid Y_1, \ldots, Y_t)$ is computed by Monte Carlo simulation. A number of paths are generated from the model with $e_t \sim N(0, \hat{\sigma}^2)$ when the normality test does not reject H_0, and the forecast is the average of these paths. Forecast intervals are taken from the empirical quantiles of the paths. The SST forecast is obtained as $\hat{X}_{t+h} = S_{t+h} + \hat{Y}_{t+h}$.

The forecast accuracy on the test data is evaluated with rolling one-step-ahead forecasts. The model parameters are fixed at the training estimates, and each forecast uses the actual data up to the previous month. The benchmark models are ARMA(2,2) as the linear part of the selected model, AR(5) as the AR model with the smallest AIC, the persistence method ($\hat{Y}_{t+1} = Y_t$), and the climatology method ($\hat{Y}_{t+1} = 0$, that is, the seasonal component only). The accuracy measures are the root mean square error (RMSE) and the mean absolute error (MAE). Accuracy on the test data is also computed separately for El Niño months, defined as months with $Y_t > 1\,^{\circ}\mathrm{C}$. In addition, a chained forecast from the end of the training data for $h = 1, \ldots, 138$ is computed to examine the long-horizon behaviour of the forecasts.

2.9 Analysis steps

The analysis consists of the following steps.

1. Split the data sequentially into training data and test data.
2. Estimate the seasonal component on the training data and determine the number of harmonics and the inclusion of the trend by BIC.
3. Test the stationarity of Y_t.
4. Identify candidate model orders from the ACF, PACF, AIC of AR models, LM test, and correlation of bilinear terms.

5. Estimate the parameters of all candidate models and compute the rolling-origin RMSE.
6. Select the model according to the criterion in Section 2.6.
7. Re-estimate the selected model and check its invertibility and the stationarity of its AR part.
8. Carry out diagnostic checks on the residuals.
9. Evaluate the forecast accuracy on the test data.
10. Re-estimate the model using all data and forecast the SST from July to December 2026.

3 Results and Discussion

3.1 Data description

During the observation period, the SST of the Niño 3 region ranges from 22.58 °C to 29.22 °C with a mean of 25.79 °C. Figure 1 shows a clear annual pattern together with interannual deviations associated with ENSO. The largest positive deviations occurred in late 1997, from late 1982 to early 1983, and in late 2015. The SST rises in these periods appear sharper than the subsequent declines, which is consistent with the asymmetry reported by Burgers and Stephenson (1999).

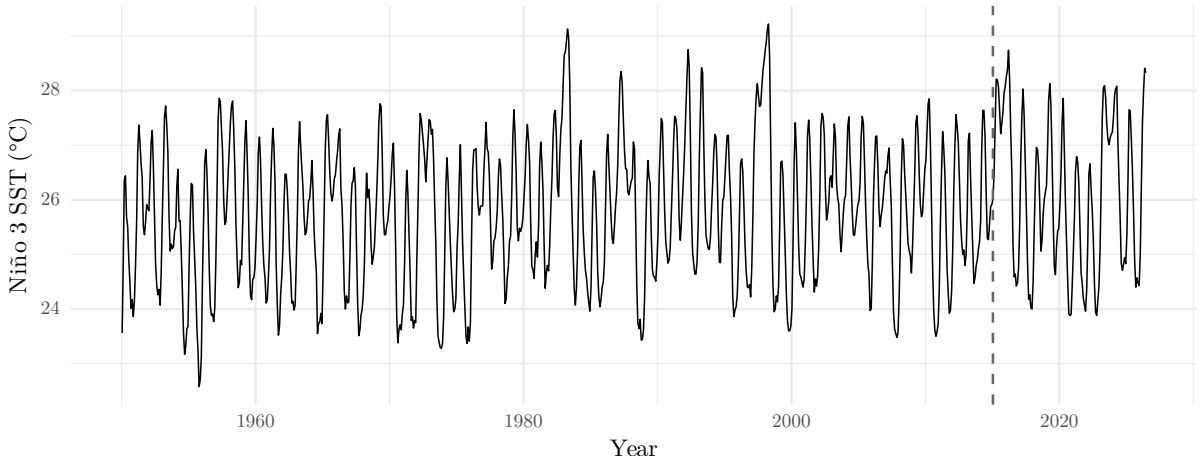


Figure 1. Monthly SST of the Niño 3 region from January 1950 to June 2026. The dashed line marks the boundary between the training data and the test data.

3.2 Seasonal component

Table 1. BIC differences from the smallest value by number of harmonics K and inclusion of the trend.

Model	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$	$K = 6$
Without trend	42.2	22.6	34.7	47.7	60.9	74.0
With trend	20.5	**0.0**	12.0	25.0	38.2	51.4

Based on Table 1, the smallest BIC is obtained for the model with $K = 2$ and a trend. For every value of K, the model with a trend has a smaller BIC than the model without a trend.

The estimated trend is 0.090 °C per decade with a standard error of 0.017. The amplitude of the annual harmonic is 1.27 °C, much larger than the amplitude of the semiannual harmonic of 0.26 °C. The seasonal component explains 53.7% of the SST variance in the training data (Figure 2), and the remaining variation is modelled through the stochastic component Y_t.

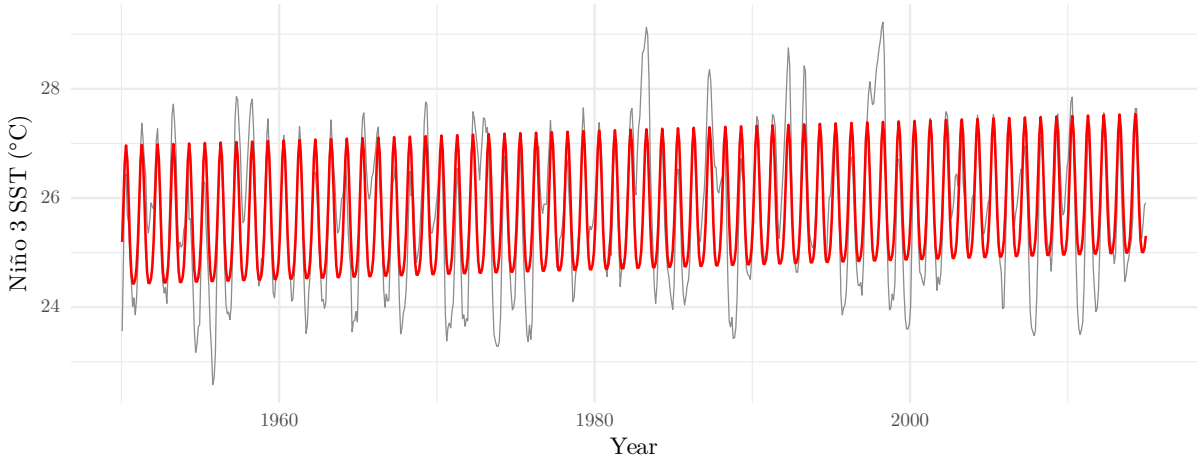


Figure 2. SST in the training data (grey) and the seasonal component with $K = 2$ and a trend (red).

3.3 Order identification

The ADF test gives a statistic of -7.64 with a p-value below 0.01, so the null hypothesis of a unit root is rejected. The KPSS test gives a statistic of 0.070 with a p-value above 0.10, so the null hypothesis of stationarity is not rejected. Both tests lead to the same conclusion that Y_t is stationary, and no differencing is required.

Figure 3 shows an ACF that decays slowly and a PACF that is significant at lags 1, 2, 4, and 5. The PACF equals 0.934 at lag 1 and -0.378 at lag 2. Based on the significant PACF lags, the candidate AR orders are $p \in \{1, 2, 4, 5\}$. The ACF and PACF patterns suggest an AR model. However, because the ACF and PACF cannot distinguish the MA part from the bilinear part (Kim 1992), the MA orders $r = 0, 1, \ldots, 12$ are still included as candidates. Among the AR(1) to AR(12) models with $\gamma = 12$, the smallest AIC is obtained for AR(5). The AIC differences of AR(4), AR(2), and AR(1) from AR(5) are 7.8, 28.4, and 167.5, respectively.

The LM test is carried out with each candidate AR order, namely AR(1), AR(2), AR(4), and AR(5), as the null model. The additional regressors are $Y_{t-i}\hat{e}_{t-1}$ for $i = 1, \ldots, 5$. The results are presented in Table 2. All null models give p-values below 0.001, so the linearity hypothesis is rejected at the 5% significance level. Hence, bilinear terms are required for every candidate AR order.

Table 2. Lagrange multiplier test (F version) against bilinear alternatives.

Null model	F	df_1	df_2	p-value
AR(1)	7.167	5	761	< 0.001
AR(2)	6.233	5	760	< 0.001
AR(4)	7.045	5	758	< 0.001
AR(5)	6.597	5	757	< 0.001

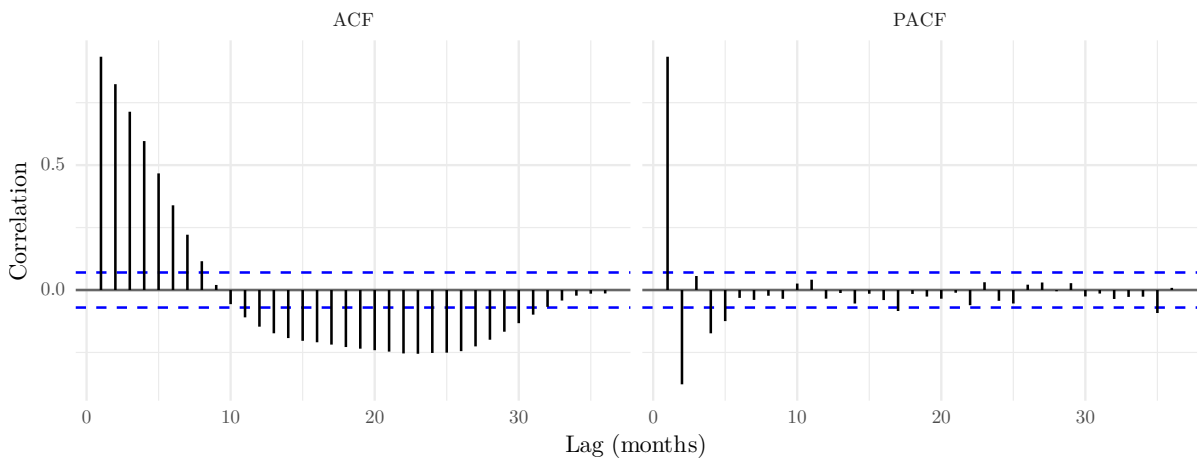


Figure 3. ACF and PACF of Y_t in the training data up to lag 36. The dashed lines show the bounds $\pm 1.96/\sqrt{n}$.

Figure 4 shows that 14 of the 144 pairs (i, j) exceed the bounds ± 0.071. The most prominent pairs lie in column $j = 1$ with $i = 1, \ldots, 5$, and the largest correlation of 0.159 occurs at the pair $(1, 1)$. All pairs in this group satisfy $i \geq j$. The other significant pairs are scattered, and some of them lie in the region $i < j$, for example $(1, 3)$ and $(2, 3)$. Of 144 tests at the 5% significance level, about seven pairs are expected to be significant by chance. Based on these results, the candidate bilinear part is restricted to $m \leq 5$ and $k \leq 3$.

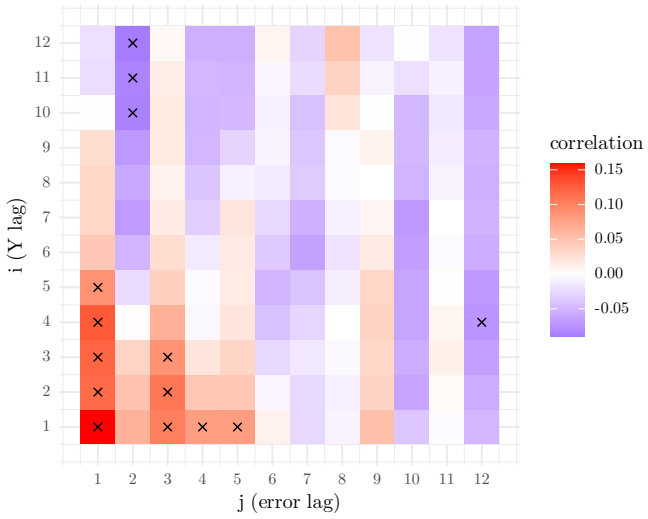


Figure 4. Correlation between the AR(5) residuals $\hat{e}_t$ and $(Y_{t-i} - \bar{Y})\hat{e}_{t-j}$. Crosses mark correlations that exceed the bounds $\pm 1.96/\sqrt{n}$.

3.4 Model selection

The combinations of $p \in \{1, 2, 4, 5\}$, $r \in \{0, \ldots, 12\}$, and 16 forms of the bilinear part give 832 candidate models. A total of 43 models required additional iterations. Of all candidates, 736 models satisfy the convergence and invertibility requirements, 91 models are not invertible, and 5 models do not converge. The models that fail these requirements are concentrated at high

MA orders. For $r = 11$ and $r = 12$, 60 of 128 models fail the requirements.

Table 3. Ten models with the smallest rolling-origin RMSE. The selected model is shown in bold.

Rank	Model	Number of parameters	AIC	RMSE (°C)
1	**BL(2,2,1,1)**	6	−2033.0	0.2420
2	BL(4,0,5,1)	10	−2028.1	0.2425
3	BL(2,2,1,3)	8	−2031.3	0.2428
4	BL(2,2,1,2)	7	−2032.3	0.2430
5	BL(2,2,2,1)	7	−2031.6	0.2431
6	BL(2,2,5,1)	10	−2029.3	0.2431
7	BL(5,0,5,1)	11	−2033.9	0.2435
8	BL(2,2,4,1)	9	−2030.1	0.2437
9	BL(2,2,3,1)	8	−2031.8	0.2440
10	BL(4,0,0,0)	5	−2000.5	0.2441

Table 3 presents the ten models with the smallest rolling-origin RMSE. The smallest RMSE of 0.2420 is obtained by the BL(2,2,1,1) model, which is therefore selected as the best model. This model contains a second-order MA part although the ACF and PACF patterns suggest an AR model. This can be explained by the ability of low-order ARMA models to approximate higher-order AR models. On the training data, the ARMA(2,2) model with five parameters has an AIC of −2009.9, smaller than that of the AR(5) model with six parameters, which has an AIC of −2008.2. Adding the bilinear term b_{11} to the ARMA(2,2) model reduces the AIC to −2033.0.

Figure 5 shows the smallest RMSE for each pair (p, r). For $p = 2$, the smallest RMSE is reached at $r = 2$, while MA orders above two do not improve the forecast accuracy. For $p = 4$ and $p = 5$, the smallest RMSE tends to increase from $r = 0$ to $r = 8$. For $r = 11$ and $r = 12$, the RMSE rises sharply above 0.31 because only one of the 16 models for each of these pairs satisfies the requirements. In addition, the AIC is not always consistent with forecast accuracy. The smallest AIC belongs to the BL(2,8,4,2) model with a value of −2043.8, but this model has an RMSE of 0.292 and ranks 716 out of 736. This indicates that additional MA and bilinear parameters can improve the fit to the training data without improving the forecast accuracy.

3.5 Parameter estimation of the BL(2,2,1,1) model

Table 4. Parameter estimates of the BL(2,2,1,1) model on the training data.

Parameter	Estimate	Standard error	z	p-value
a_1	−1.7161	0.0482	−35.59	< 0.001
a_2	0.7520	0.0425	17.70	< 0.001
c_1	−0.3523	0.0589	−5.98	< 0.001
c_2	−0.2564	0.0460	−5.58	< 0.001
b_{11}	0.2187	0.0368	5.95	< 0.001
α	−0.0134	0.0045	−2.97	0.003

The parameter estimates are presented in Table 4. Estimation with staged starting values and with zero starting values gives the same AIC of −2033.02 and converges in 13 iterations.

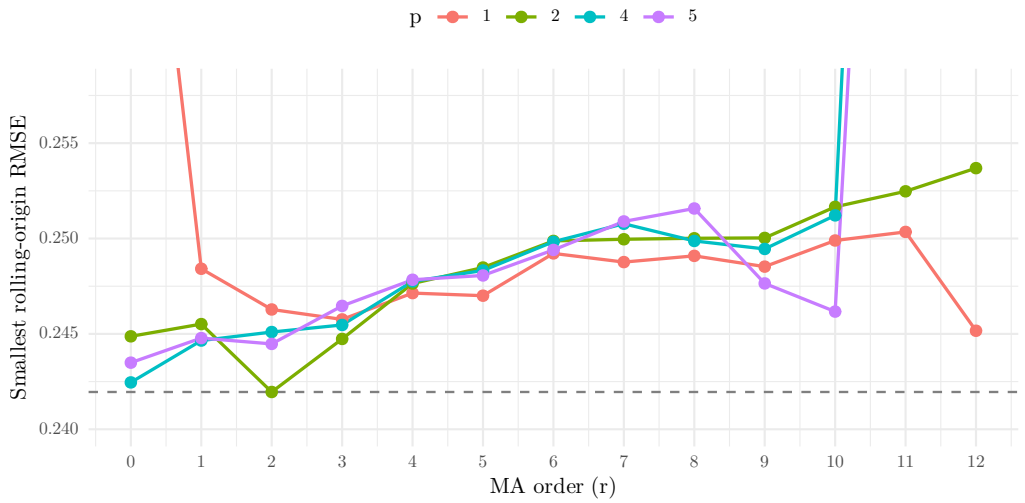


Figure 5. Smallest rolling-origin RMSE for each pair (p, r). The dashed line shows the smallest value among all models. The vertical axis is truncated at 0.258.

The estimated model is

$$\hat{Y}_t = 1.716\, Y_{t-1} - 0.752\, Y_{t-2} - 0.013 - 0.352\, e_{t-1} - 0.256\, e_{t-2} + 0.219\, Y_{t-1} e_{t-1},$$

with $\hat{\sigma}^2 = 0.0698$.

According to the check in equation (3), the model is invertible. The roots of the AR polynomial $1 + a_1 z + a_2 z^2$ form a pair of complex numbers with modulus 1.153, while the roots of the MA polynomial $1 + c_1 z + c_2 z^2$ have moduli 1.404 and 2.778. All moduli are greater than one, so the AR part is stationary and the MA part is invertible. The complex AR roots correspond to a damped cycle with a period of about 43 months or 3.6 years. This period is consistent with ENSO, which recurs every few years (Timmermann et al. 2018).

All parameters are significant at the 5% level. The standard errors of the bilinear coefficient are computed from the approximation $\hat{\sigma}^2 (J'J)^{-1}$, which may underestimate them, so the p-value of b_{11} should be interpreted with caution. The effective coefficient of the error e_{t-1} is $c_1 + b_{11} Y_{t-1} = -0.352 + 0.219\, Y_{t-1}$. This coefficient is negative when $Y_{t-1} < 1.61$ °C, so the previous error is damped, and positive when the previous anomaly exceeds 1.61 °C, so a positive error is amplified. This mechanism is consistent with strong El Niño events developing more sharply than La Niña events.

3.6 Diagnostic checking

Table 5. Diagnostic checks of the BL(2,2,1,1) residuals on the training data.

Test	Statistic	p-value
Ljung-Box on residuals (lag 12)	19.392	0.013
McLeod-Li on squared residuals (lag 12)	13.864	0.309
Skewness u_1	−1.471	0.141
Kurtosis u_2	1.246	0.213
Normality $u_1^2 + u_2^2$	3.716	0.156

The diagnostic results are presented in Table 5 and Figure 6. The normality test does not reject H_0, with a p-value of 0.156, a skewness of -0.133, and an excess kurtosis of 0.220, so the Monte Carlo simulation uses normally distributed errors. The McLeod-Li test does not reject H_0, with a p-value of 0.309, so there is no dependence in the squared residuals. The Ljung-Box test rejects H_0 at the 5% level, with a p-value of 0.013. The residual ACF in Figure 6 shows that the remaining autocorrelation is small, although the model does not fully capture the linear autocorrelation structure.

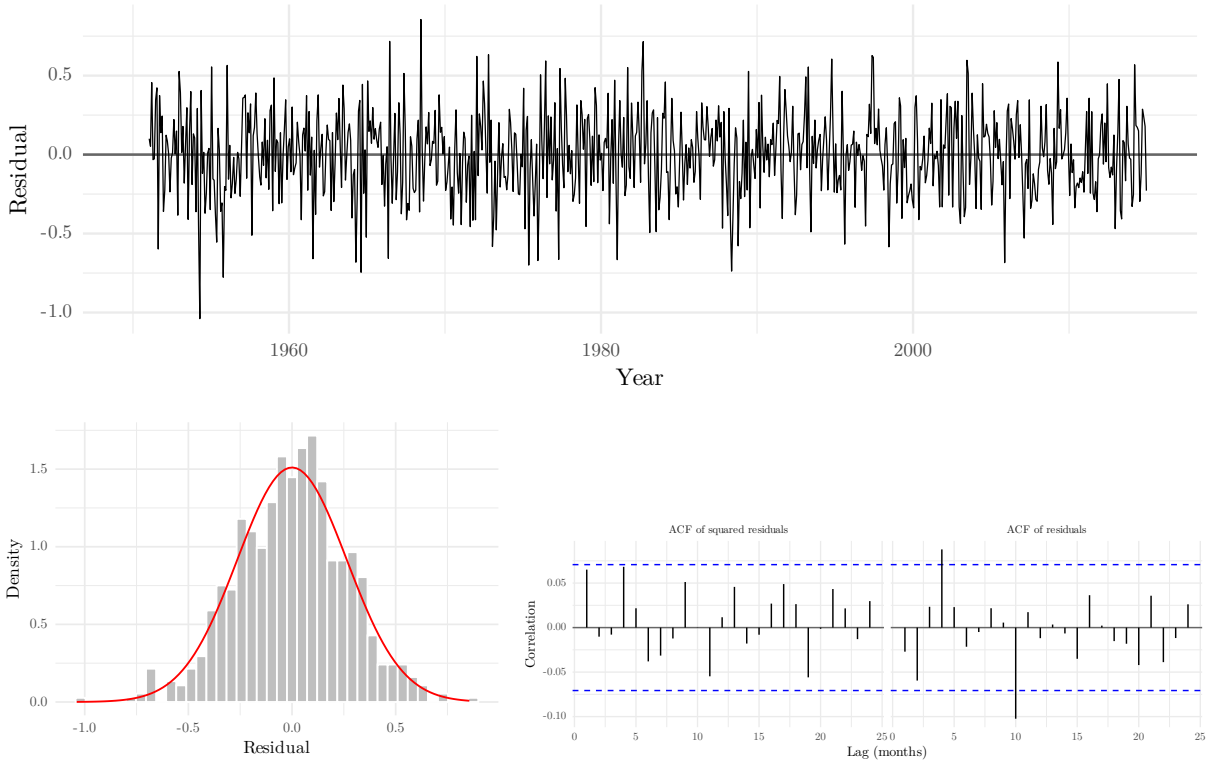


Figure 6. Residuals of the BL(2,2,1,1) model on the training data. The top panel shows the residuals over time, the bottom left panel shows the residual histogram with the $N(0, \hat{\sigma}^2)$ density (red), and the bottom right panel shows the ACF of the residuals and of the squared residuals.

3.7 Forecast evaluation on the test data

Figure 7 presents the fitted values on the training data and the chained forecasts for $h = 1, \ldots, 138$ from December 2014. The fitted values follow the actual data closely. The long-horizon forecasts return to the seasonal component within a few months because Y_t is stationary with a mean close to zero. Consequently, the RMSE of the long-horizon forecasts of 0.986 °C is almost equal to the RMSE of the seasonal component of 0.980 °C. A total of 95.7% of the test data lie within the 95% forecast interval. This shows that forecasts from a single origin are informative only for the first few months.

The accuracy of the rolling one-step-ahead forecasts is presented in Table 6. The BL(2,2,1,1) model gives an RMSE of 0.2580 °C. This value is 17% smaller than that of the persistence method, 74% smaller than that of the climatology method, and 1.7% smaller than that of the ARMA(2,2) model, which is its linear part. Compared with the AR(5) model, the RMSE of the BL(2,2,1,1) model is 0.6% larger, so the accuracy of the two models can be regarded as

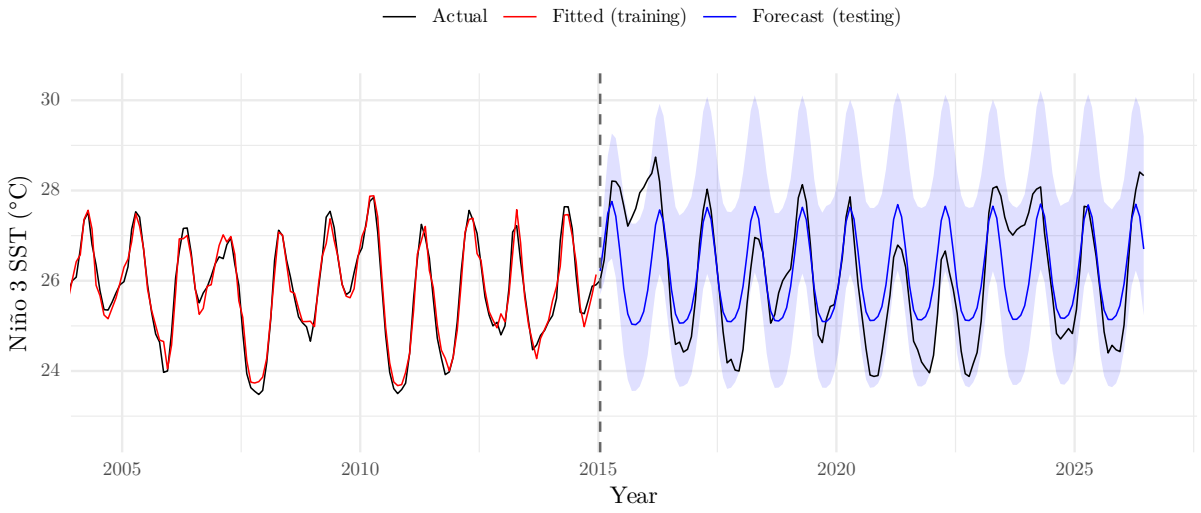


Figure 7. Actual data (black), fitted values on the training data (red), and chained forecasts on the test data (blue) with the 95% forecast interval (blue band), 2005 to 2026.

Table 6. Accuracy of rolling one-step-ahead forecasts on the test data (°C). The El Niño column uses 21 months with $Y_t > 1\,^{\circ}\mathrm{C}$.

	All months		RMSE by condition	
Method	RMSE	MAE	El Niño	Other
BL(2,2,1,1)	0.2580	0.2095	0.2731	0.2552
ARMA(2,2)	0.2625	0.2133	0.2968	0.2559
AR(5)	0.2564	0.2097	0.2868	0.2506
Persistence	0.3117	0.2535		
Climatology	0.9804	0.7628		

comparable. Figure 8 shows that the one-step-ahead forecasts follow the actual data closely, including the El Niño peaks of 2015 to 2016 and 2023 to 2024.

A clearer difference appears in El Niño months, namely June 2015 to March 2016, June 2023 to February 2024, and May and June 2026. In these months, the BL(2,2,1,1) model gives an RMSE of 0.2731 °C, which is 8.0% smaller than that of the ARMA(2,2) model and 4.8% smaller than that of the AR(5) model. In the other months, by contrast, the AR(5) model gives a slightly smaller RMSE. These results indicate that the bilinear term matters mainly when SST anomalies are large, in line with the effective coefficient $c_1 + b_{11}Y_{t-1}$, which depends on the anomaly of the previous month.

3.8 Forecast for July to December 2026

The forecast is made after re-estimating the seasonal component and the BL(2,2,1,1) model using all 918 observations. The re-estimated model converges and is invertible, and its coefficients are close to the training estimates. The coefficient b_{11} changes from 0.219 to 0.204, while c_1 changes from -0.352 to -0.353. The residual normality test does not reject H_0, with a p-value of 0.210, so the simulation uses normally distributed errors with 20000 paths. The Ljung-Box test again rejects H_0, with a p-value < 0.001.

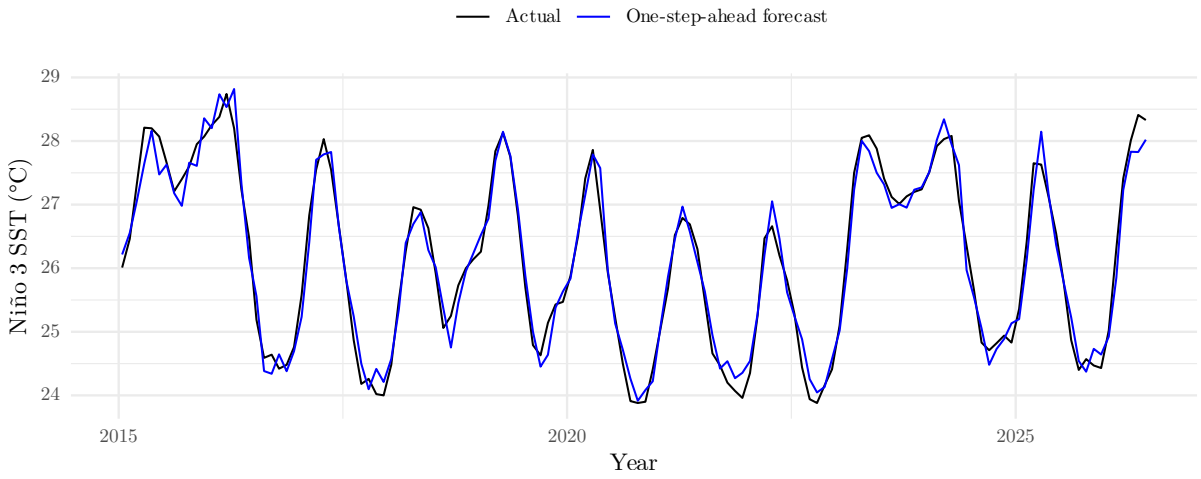


Figure 8. Actual data (black) and rolling one-step-ahead forecasts of the BL(2,2,1,1) model (blue) on the test data.

Table 7. Forecasts of the Niño 3 SST for July to December 2026 (°C).

Month	Seasonal component	Anomaly $\hat{Y}$	SST forecast	95% interval	Naive forecast	ARMA(2,2)
July	25.87	1.92	**27.79**	27.28 to 28.31	27.79	27.65
August	25.32	1.97	**27.29**	26.28 to 28.38	27.27	27.02
September	25.11	1.94	**27.05**	25.64 to 28.60	27.01	26.69
October	25.11	1.85	**26.95**	25.25 to 28.95	26.88	26.56
November	25.19	1.71	**26.90**	24.97 to 29.30	26.79	26.49
December	25.40	1.55	**26.95**	24.86 to 29.69	26.81	26.54

The forecasts are presented in Table 7 and Figure 9. In June 2026, the observed SST was 28.33 °C with Y_t equal to 1.66 °C. The forecast anomaly rises to 1.92 °C in July and 1.97 °C in August and then declines to 1.55 °C in December. Together with the decline of the seasonal component, the SST forecast decreases from 27.79 °C in July to 26.90 °C in November and 26.95 °C in December. These results indicate that warm conditions are expected to persist until the end of 2026 with weakening intensity.

The forecast interval widens as the horizon increases. The 95% forecast interval for December is 24.86 to 29.69 °C and extends further upward, in line with the asymmetric nature of the bilinear model. The Monte Carlo forecasts are 0.00 to 0.14 °C higher than the naive forecasts because the naive forecasts ignore the positive expected value of the term $b_{11}Y_{t-1}e_{t-1}$. The ARMA(2,2) model produces forecasts that decline faster, reaching 26.54 °C in December, while the BL(2,2,1,1) model keeps the positive anomaly for longer. Since forecast accuracy decreases with the horizon, the forecasts for the last months should be interpreted together with their forecast intervals.

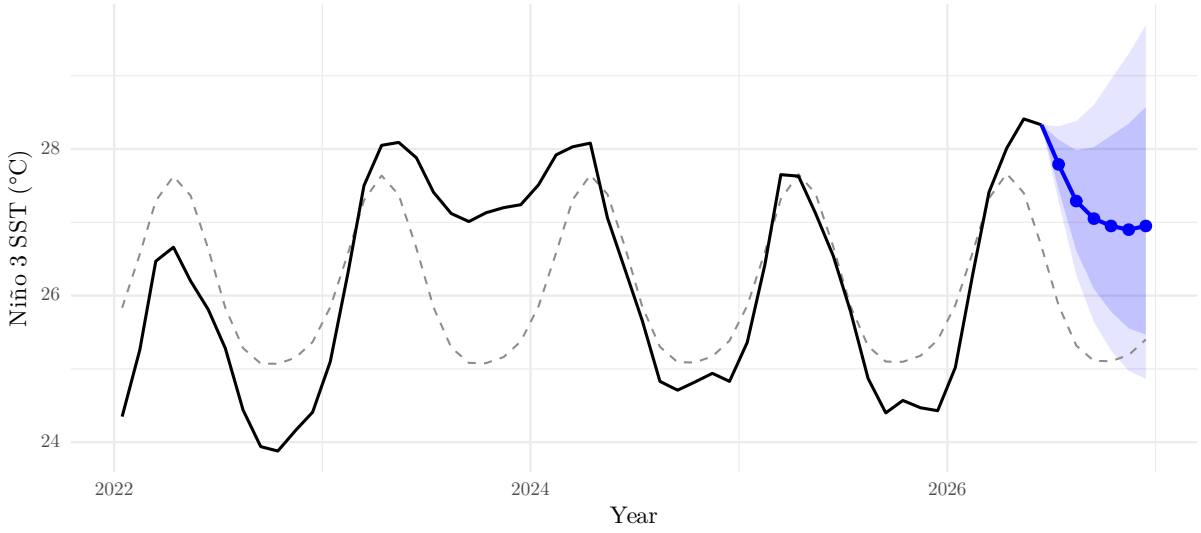


Figure 9. Actual SST of the Niño 3 region since 2022 (black), seasonal component (grey dashed), and forecasts for July to December 2026 (blue) with 80% and 95% forecast intervals.

4 Conclusions and Recommendations

4.1 Conclusions

The seasonal pattern of the monthly SST of the Niño 3 region can be represented by two Fourier harmonics and a linear trend of 0.09 °C per decade, while its stochastic component is stationary and nonlinear according to the Lagrange multiplier test. Of the 832 models evaluated, the BL(2,2,1,1) model was selected as the model with the smallest rolling-origin RMSE. MA orders above two do not improve the forecast accuracy, and the smallest AIC does not point to the model with the best forecast accuracy. On the test data, the one-step-ahead forecast RMSE of the BL(2,2,1,1) model is 0.258 °C, which is smaller than that of the ARMA(2,2) model, the persistence method, and the climatology method, and comparable to that of the AR(5) model. In El Niño months, the BL(2,2,1,1) model yields the smallest RMSE among all benchmark models. The forecast based on the full data shows a positive anomaly that peaks at 1.97 °C in August 2026 and then weakens to 1.55 °C in December 2026.

4.2 Recommendations

Future studies are recommended to compute the standard errors of the bilinear coefficients using the bootstrap method.

References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716-723.

Bera, A. K. and Higgins, M. L. (1993). *ARCH and Bilinearity as Competing Models for Nonlinear Dependence*. BEBR Faculty Working Paper 93-0116, University of Illinois at Urbana-Champaign.

Brown, B. W. and Mariano, R. S. (1989). Predictors in dynamic nonlinear models: Large-sample behavior. *Econometric Theory*, 5(3), 430-452.

Burgers, G. and Stephenson, D. B. (1999). The "normality" of El Niño. *Geophysical Research Letters*, 26, 1027-1030.

Chen, C. W. S., Cherng, T. H., and Wu, B. (2001). On the selection of subset bilinear time series models: A genetic algorithm approach. *Computational Statistics*, 16(4), 505-517.

De Livera, A. M., Hyndman, R. J., and Snyder, R. D. (2011). Forecasting time series with complex seasonal patterns using exponential smoothing. *Journal of the American Statistical Association*, 106(496), 1513-1527.

Dickey, D. A. and Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74, 427-431.

Gabr, M. M. and Subba Rao, T. (1981). The estimation and prediction of subset bilinear time series models with applications. *Journal of Time Series Analysis*, 2(3), 155-171.

Granger, C. W. J. and Andersen, A. P. (1978a). *An Introduction to Bilinear Time Series Models*. Vandenhoeck & Ruprecht, Göttingen.

Granger, C. W. J. and Andersen, A. P. (1978b). On the invertibility of time series models. *Stochastic Processes and their Applications*, 8(1), 87-92.

Huang, B., Thorne, P. W., Banzon, V. F., Boyer, T., Chepurin, G., Lawrimore, J. H., Menne, M. J., Smith, T. M., Vose, R. S., and Zhang, H.-M. (2017). Extended Reconstructed Sea Surface Temperature, version 5 (ERSSTv5): Upgrades, validations, and intercomparisons. *Journal of Climate*, 30(20), 8179-8205.

Kim, W. K. (1992). Autocorrelation function structure of bilinear time series models. *Journal of the Korean Statistical Society*, 21(1), 47-58.

Kumar, K. (1986). On the identification of some bilinear time series models. *Journal of Time Series Analysis*, 7(2), 117-122.

Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., and Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1-3), 159-178.

Li, W. K. (1984). On the autocorrelation structure and identification of some bilinear time series. *Journal of Time Series Analysis*, 5(3), 173-181.

Ljung, G. M. and Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65, 297-303.

Lomnicki, Z. (1961). Tests for departure from normality in the case of linear stochastic processes. *Metrika*, 4(1), 37-62.

McLeod, A. I. and Li, W. K. (1983). Diagnostic checking ARMA time series models using squared-residual autocorrelations. *Journal of Time Series Analysis*, 4, 269-273.

Said, S. E. and Dickey, D. A. (1984). Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika*, 71, 599-607.

Saikkonen, P. and Luukkonen, R. (1988). Lagrange multiplier tests for testing non-linearities in time series models. *Scandinavian Journal of Statistics*, 15, 55-68.

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461-464.

Subba Rao, T. (1981). On the theory of bilinear time series models. *Journal of the Royal Statistical Society, Series B*, 43(2), 244-255.

Subba Rao, T. and Gabr, M. M. (1984). *An Introduction to Bispectral Analysis and Bilinear Time Series Models*. Lecture Notes in Statistics 24. Springer, New York.

Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4), 437-450.

Teräsvirta, T. (1994). Testing linearity and modelling nonlinear time series. *Kybernetika*, 30(3), 319-330.

Timmermann, A., et al. (2018). El Niño Southern Oscillation complexity. *Nature*, 559, 535-545.

Trenberth, K. E. (1997). The definition of El Niño. *Bulletin of the American Meteorological Society*,

78(12), 2771-2777.

Wu, B. and Shih, N.-H. (1992). On the identification problem for bilinear time series models. *Journal of Statistical Computation and Simulation*, 43(3-4), 129-161.